How GrepQA Works Now (v21)

Core Thesis

Use LLM-generated grep predicates instead of embedding retrieval.

1. Generate 3 predicates from different question-side angles
2. Retrieve with `top_k=8` and context `window=2`
3. Answer with temporal reasoning over session dates

```
"""
GrepQA v21:
3 predicates, all question-side
(no answer guessing)
"""
```

```
Predicate 1: most specific word
Predicate 2: different word
Predicate 3: broader context / synonym
execute(top_k=8, window=2)
```

GrepQA Results Main Table

Method	Single	Temp.	Multi	Open	Overall
no_context*	12.34	15.49	21.15	15.22	15.12
full_context*	34.26	36.20	25.67	49.38	42.39
GrepQA*	18.92	38.29	24.26	31.03	29.90
GrepQA v2*	20.20	42.79	26.70	38.26	35.18
GrepQA v6*	28.78	48.96	30.30	53.73	46.71
GrepQA v21*	**36.04**	**54.29**	**31.77**	**58.22**	**51.69**
oracle*	99.21	97.51	98.23	99.93	99.19
LoCoMo	25.02	18.41	12.04	40.36	31.21
ReadAgent	9.15	12.60	5.31	9.67	9.91
MemoryBank	5.00	9.68	5.56	6.61	6.89
MemGPT	26.65	25.52	9.15	41.04	33.18
A-Mem	27.02	45.85	12.14	44.65	39.65
LangMem	35.51	30.75	26.04	40.91	36.88
Zep	35.74	42.00	19.37	49.56	43.57
OpenAI Memory	34.30	14.04	20.09	39.31	31.93
Mem0	38.72	48.93	28.64	47.65	45.10
Mem0^g	38.09	51.55	24.32	49.27	46.14

(* = my runs, others from Mem0 paper)

Version Changelog

Version	Overall F1	Key Change
v1	29.90	Baseline: single keyword predicate, top-k retrieval
v2	35.18	Added sliding window context expansion and multi-query retrieval
v3	28.96	Precision-focused retrieval: window=0, top-k=2, reranking, smarter fallback
v4	22.43	Adaptive top-k and stricter answer extraction prompt
v5	31.51	Selective LLM filtering when many turns are retrieved
v6	44.69	New baseline: 2 predicates, must/should scoring, speaker filter
v7*	41.62	Added session-date-aware time_hint scoring
v8*	35.18	Added answer-type-aware answering prompts
v9*	44.57	Switched to 3 question-side predicates
v10	46.46	Added date-aware answer prompt on top of v6-k8
v11	46.16	Answer prompt: include all values for list questions
v12	47.51	Balanced answer prompt: complete but concise
v13	48.10	Resolve relative time expressions using session dates
v14	47.46	Combined temporal reasoning with "be complete" instruction
v15*	45.26	Added answer-anticipation words into predicate generation
v16*	47.41	Added conservative third answer-anticipation predicate
v17*	44.06	Second predicate explicitly targets answer-turn vocabulary
v18	48.16	Disabled speaker fallback to reduce noisy retrieval
v19*	46.48	Reduced top-k from 8 to 5 to cut retrieval noise
v20*	45.69	Switched to 3 predicates, including answer-side keyword guessing
v21	**51.69**	3 predicates; without answer-side keyword guessing

The gains did not come from one trick. They came from iterative systems search over predicates, context windows, answer prompts, and retrieval noise control.